

EN55022B Compliant 40V, Dual 4A or Single 8A Step-Down or 50W Inverting µModule Regulator

FEATURES

- Dual 4A/Single 8A Low EMI Switch Mode Power Supply
- EN55022 Class B Compliant
- Two Fully Independent Channels, Each Configurable for Positive or Negative Output Voltage Polarity
- Output Voltage Range: $0.5V \leq |V_{OUTn}^+ - V_{OUTn}^-| \leq 26.5V$
- Wide Input Voltage Range: Up to 40V
 - 3.1V or 3.6V Start-Up, Configuration-Dependent
- $\pm 1.67\%$ Total DC Output Voltage Error Over Line, Load and Temperature
- Analog Output Current Indicator (Positive- V_{OUT} Only)
- LDO_{OUT}: 5V Fixed, 25mA Capable LDO
- Parallelable with LTM4651/LTM4653
- Constant-Frequency Current Mode Control
- Power Good Indicators and Programmable Soft-Start
- Overcurrent and Overtemperature Protection
- 16mm × 16mm × 5.01mm BGA Package

APPLICATIONS

- Automated Test and Measurement
- Avionics and Industrial Control Systems
- Video, Imaging and Instrumentation

DESCRIPTION

The LTM®4655 is an ultralow noise 40V, dual 4A or single 8A DC/DC µModule® regulator designed to meet the radiated emissions requirements of EN55022. Its channels are fully independent, parallelable and capable of delivering positive or negative output polarity. Conducted emission requirements can be met by adding standard filter components. Included in the package are the switching controllers, power MOSFETs, inductors, filters and support components. A 5V, 25mA LDO and clock generator enable phase interleaving of the power switching stages, for improved EMC performance.

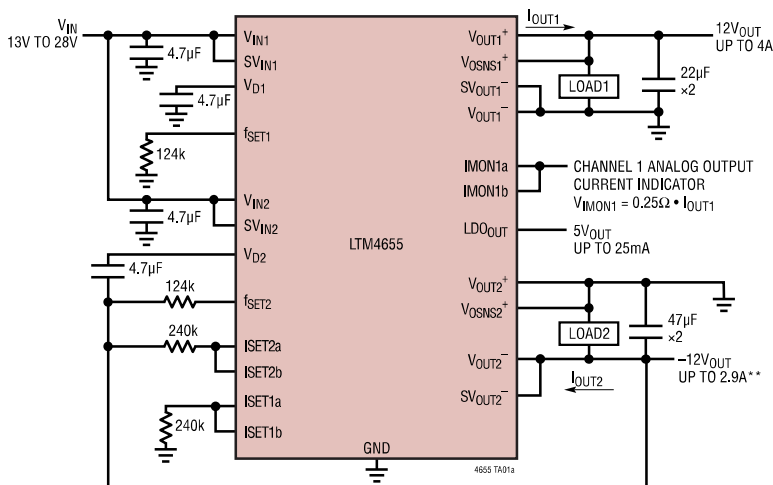
The LTM4655 can regulate positive V_{OUTn}^+ voltages between 0.5V and 26.5V from a 3.1V to 40V input. The LTM4655 can regulate negative V_{OUTn}^- voltages between -0.5V and -26.5V from a maximum input range of 3.6V to 40V, with the span from V_{INn} to V_{OUTn}^- not to exceed 40V. A switching frequency range of 250kHz to 3MHz is supported.

The LTM4655 is offered in a 16mm × 16mm × 5.01mm BGA package with SnPb or RoHS compliant terminal finish.

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TYPICAL APPLICATION

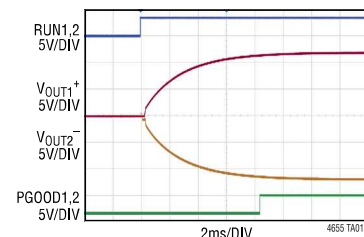
Concurrent, $\pm 12V$ Output DC/DC µModule Regulator*



* FOR COMPLETE CIRCUIT, SEE FIGURE 50.

** FOR CHANNELS CONFIGURED TO REGULATE NEGATIVE V_{OUTn}^- : CURRENT LIMIT FREQUENCY-FOLDBACK INCEPTION IS A FUNCTION OF V_{INn} , V_{OUTn}^- , AND f_{SWn} . CONTINUOUS OUTPUT CURRENT CAPABILITY IS SUBJECT TO DETAILS OF APPLICATION IMPLEMENTATION. SEE NOTES 2 AND 3 AND THE APPLICATIONS INFORMATION SECTION, FOR DETAILS.

Output Voltage Start-Up Waveforms



ORDER INFORMATION

PART NUMBER	PAD OR BALL FINISH	PART MARKING*		PACKAGE TYPE	MSL RATING	TEMPERATURE RANGE (SEE NOTE 2)
		DEVICE	FINISH CODE			
LTM4655EY#PBF	SAC305 (RoHS)	LTM4655Y	e1	BGA	3	–40°C to 125°C
LTM4655IY#PBF	SAC305 (RoHS)	LTM4655Y	e1	BGA	3	–40°C to 125°C
LTM4655MPY#PBF	SAC305 (RoHS)	LTM4655Y	e1	BGA	3	–55°C to 125°C
LTM4655IY	SnPb (63/37)	LTM4655Y	e0	BGA	3	–40°C to 125°C
LTM4655MPY	SnPb (63/37)	LTM4655Y	e0	BGA	3	–55°C to 125°C

• Contact the factory for parts specified with wider operating temperature ranges. *Pad or ball finish code is per IPC/JEDEC J-STD-609.

• [Recommended LGA and BGA PCB Assembly and Manufacturing Procedures](#)

• [LGA and BGA Package and Tray Drawings](#)

ELECTRICAL CHARACTERISTICS The ● denotes the specifications which apply over the specified internal operating temperature range (Note 2). Specified as each individual output channel (Note 5). $T_A = 25^\circ\text{C}$, Test Circuit 1 (positive- V_{OUT} , noninverting step-down configuration with $V_{OUTn} = \text{GND}$), $V_{INn} = \text{SV}_{INn} = 36\text{V}$, $\text{EXTV}_{CCn} = 24\text{V}$, $\text{RUNn} = 3.3\text{V}$, $\text{R}_{ISETn} = 480\text{k}$, $\text{R}_{ISETn}^+ = 57.6\text{k}\Omega$, $\text{f}_{\text{SW}n} = 1.5\text{MHz}$ (CLKINn driven with 1.5MHz clock signal) and voltages referred to GND unless otherwise noted.

SYMBOL	PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
$\text{SV}_{INn(\text{DC})}$, $\text{V}_{INn(\text{DC})}$	Input DC Voltage in Positive- V_{OUT} Configuration	$V_{OUTn}^- = \text{GND}$	● 3.1		40	V
$\text{V}_{OUTn(\text{RANGE})}^+$	Range of Positive Output Voltage Regulation	$0.5\text{V} \leq \text{ISETna} - \text{SV}_{OUTn}^- \leq 26.5\text{V}$, $\text{I}_{OUTn}^+ = 0\text{A}$ (See Note 7)	● 0.5		26.5	V
$\text{V}_{OUTn(24\text{VDC})}^+$	Output Voltage Total Variation with Line and Load at $\text{V}_{OUTn}^+ = 24\text{V}$	$29\text{V} \leq \text{V}_{INn} \leq 40\text{V}$, $0\text{A} \leq \text{I}_{OUTn}^+ \leq 4\text{A}$, $\text{C}_{\text{IN}Hn} = 4.7\mu\text{F}$, $\text{C}_{\text{D}n} = 4.7\mu\text{F}$, $\text{C}_{\text{OUT}Hn} = 2 \times 47\mu\text{F}$, CLKINn Driven with 1.5MHz Clock	● 23.6	24	24.4	V
$\text{V}_{OUTn(0.5\text{VDC})}^+$	Output Voltage Total Variation with Line and Load at $\text{V}_{OUTn}^+ = 0.5\text{V}$	Measuring $\text{V}_{\text{OSNS}n}^+$ to ISETna $3.1\text{V} \leq \text{V}_{INn} \leq 13.2\text{V}$, $0\text{A} \leq \text{I}_{OUTn}^+ \leq 4\text{A}$, $\text{C}_{\text{IN}Hn} = 4.7\mu\text{F}$, $\text{C}_{\text{D}n} = 4.7\mu\text{F}$, $\text{C}_{\text{OUT}Hn} = 2 \times 47\mu\text{F}$, $\text{ISETna} = 500\text{mV}$, $\text{R}_{\text{ISET}n} = \text{N/U}$ (Note 6)	● –15	0	15	mV
$\text{R}_{\text{SVIN}Fn}$	Resistor Between SV_{INn} and $\text{SV}_{\text{IN}Fn}$			1		Ω

Input Specifications

$\text{V}_{INn(\text{UVLO})}$	SV_{INn} Undervoltage Lockout Threshold	SV_{INn} Rising SV_{INn} Falling Hysteresis	● 2.4 ● 150	2.85 2.6 250	3.1 2.9	V V mV
$\text{I}_{\text{INRUSH}}(\text{VIN}n)$	Input Inrush Current at Start-Up	$\text{C}_{\text{IN}Hn} = 4.7\mu\text{F}$, $\text{C}_{\text{D}n} = 4.7\mu\text{F}$, $\text{C}_{\text{OUT}Hn} = 2 \times 47\mu\text{F}$; $\text{I}_{OUTn}^+ = 0\text{A}$, ISETna Electrically Connected to ISETnb		300		mA
$\text{I}_{\text{Q}}(\text{SVIN}n)$	Input Supply Bias Current	Shutdown, $\text{RUNn} = \text{GND}$ $\text{RUNn} = 3.3\text{V}$		16 450	30	μA μA
$\text{I}_{\text{S}}(\text{VIN}n)$	Input Supply Current	CLKINn Open Circuit, $\text{I}_{OUTn}^+ = 4\text{A}$		2.9		A
$\text{I}_{\text{S}}(\text{VIN}n, \text{SHUTDOWN})$	Input Supply Current in Shutdown	Shutdown, $\text{RUNn} = \text{GND}$		4		μA

Output Specifications

I_{OUTn}^+	V_{OUTn}^+ Output Continuous Current Range	(Note 3)		0	4	A
$\Delta \text{V}_{OUTn(\text{LINE})}^+ / \text{V}_{OUTn}^+$	Line Regulation Accuracy	$\text{I}_{OUTn}^+ = 0\text{A}$, $29\text{V} \leq \text{V}_{INn} \leq 40\text{V}$	●	0.05	0.1	%
$\Delta \text{V}_{OUTn(\text{LOAD})}^+ / \text{V}_{OUTn}^+$	Load Regulation Accuracy	$\text{V}_{INn} = 36\text{V}$, $0\text{A} \leq \text{I}_{OUTn}^+ \leq 4\text{A}$	●	0.05	0.75	%
$\text{V}_{OUTn(\text{AC})}^+$	Output Voltage Ripple, V_{OUTn}^+	$\text{V}_{INn} = 12\text{V}$, $\text{ISETna} = 5\text{V}$		2		mV _{P-P}

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SYMBOL	PARAMETER	CONDITIONS		MIN	TYP	MAX	UNITS
f_{SN}	V_{OUTn}^+ Ripple Frequency	$R_{ISETn} = 57.6\text{k}\Omega$, CLKINn Open Circuit	●	1.7	1.95	2.2	MHz
$\Delta V_{OUTn(\text{START})}^+$	Turn-On Overshoot				8		mV
t_{STARTn}	Turn-On Start-Up Time	Delay Measured from V_{INn} Toggling from 0V to 36V to PG00Dn Exceeding 3V; PG00Dn. Having a 100k Ω Pull-Up to 3.3V with Respect to GND, VPGFBn Resistor Divider Network as Shown in Test Circuit 1, $R_{ISETna} = 480\text{k}\Omega$ and ISETna Electrically Connected to ISETnb and CLKIN Driven with 1.5MHz Clock	●		4	9	ms
$\Delta V_{OUTn(\text{LS})}^+$	Peak Output Voltage Deviation for Dynamic Load Step	I_{OUTn}^+ : 0A to 2A and 2A to 0A Load Steps in 1 μs , $C_{OUTHn} = 47\mu\text{F} \times 2$			400		mV
t_{SETTLEn}	Settling Time for Dynamic Load Step	I_{OUTn}^+ : 0A to 2A and 2A to 0A Load Steps in 1 μs , $C_{OUTHn} = 47\mu\text{F} \times 2$			50		μs
$I_{OUTn(\text{OCL})}^+$	I_{OUTn}^+ Output Current Limit				5.5		A

Control Section

I_{ISETna}	Reference Current of ISETna Pin	$V_{ISETna} = 0.5\text{V}$, $3.1\text{V} \leq V_{INn} \leq 13.2\text{V}$ $V_{ISETna} = 24\text{V}$, $29\text{V} \leq V_{INn} \leq 40\text{V}$	● ●	49.3 49	50 50	50.7 51	μA μA
I_{VOSNSn}^+	V_{OSNSn}^+ Leakage Current	$V_{VOSNSn}^+ = 28\text{V}$			290		μA
$t_{\text{ONn(MIN)}}$	Minimum On-Time	(Note 4)			60		ns
V_{RUNn}	RUNn Turn-On/-Off Thresholds	RUNn Input Turn-On Threshold, RUNn Rising RUNn Hysteresis	●	1.08	1.2 130	1.32	V mV
I_{RUNn}	RUNn Leakage Current	$\text{RUNn} = 3.3\text{V}$	●		0.1	50	nA

Oscillator and Phase-Locked Loop (PLL)

f_{OSCn}	Oscillator Frequency Accuracy	$V_{INn} = 12\text{V}$, $I_{SETna} = 5\text{V}$, and: f_{SETn} Open-Circuit $R_{ISETn} = 57.6\text{k}\Omega$ (See f_{SN} Specification)	●	360	400 1.95	440	kHz MHz
f_{SYNCn}	PLL Synchronization Capture Range	$V_{INn} = 12\text{V}$, $I_{SETna} = 5\text{V}$, CLKINn Driven with a GND Referred Clock Toggling from 0.4V to 1.2V and Having a Clock Duty Cycle: From 10% to 90%; f_{SETn} Open Circuit From 40% to 60%; $R_{ISETn} = 57.6\text{k}\Omega$		250 1.3		550 3	kHz MHz
V_{CLKINn}	CLKINn Input Threshold	V_{CLKINn} Rising V_{CLKINn} Falling		1.2		0.4	V V
I_{CLKINn}	CLKINn Input Current	$V_{\text{CLKINn}} = 5\text{V}$ $V_{\text{CLKINn}} = 0\text{V}$		-20	230 -5	500	μA μA

Power Good Feedback Input and Power Good Output

$\text{OV}_{\text{PGDFBn}}$	Output Overvoltage PG00Dn Upper Threshold	PGDFBn Rising	●	620	645	675	mV
$\text{UV}_{\text{PGDFBn}}$	Output Undervoltage PG00Dn Lower Threshold	PGDFBn Falling	●	525	555	580	mV
ΔV_{PGDFBn}	PG00Dn Hysteresis	PGDFBn Returning			8		mV
R_{PGDFBn}	Resistor Between PGDFB1n and $\text{SV}_{\text{OUTn}}^-$			4.94	4.99	5.04	k Ω
R_{PG00Dn}	PG00Dn Pull-Down Resistance	$V_{\text{PG00Dn}} = 0.1\text{V}$, $V_{\text{PGDFBn}} < \text{UV}_{\text{PGDFBn}}$ or $V_{\text{PGDFBn}} > \text{OV}_{\text{PGDFBn}}$			700	1500	Ω
$I_{\text{PG00Dn(LEAK)}}$	PG00Dn Leakage Current	$V_{\text{PG00Dn}} = 3.3\text{V}$, $\text{UV}_{\text{PGDFBn}} < V_{\text{PGDFBn}} < \text{OV}_{\text{PGDFBn}}$			0.1	1	μA